

Person Identification System using Deep Learning Architectures

A Study on Human Recognition using Convolutional Neural Networks

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Abstract

Identifying individuals is an important task in computer vision. It is used in many areas such as surveillance systems, biometric verification, and smart access control. This project proposes a deep learning method for recognizing individuals with Convolutional Neural Networks (CNNs) and transfer learning.

We created a customized dataset with 12 subjects and 1,000 images for each class, totaling 12,000 images. The dataset shows variations in lighting, facial expressions, and backgrounds, making it suitable for real-world situations. We trained and evaluated several models, including a baseline CNN, MobileNetV2, EfficientNet, and VGG16, using a consistent approach.

Our experimental results show that the baseline CNN achieves an accuracy of 91.57%. Transfer learning models perform much better. Notably, MobileNetV2 reaches 100% validation accuracy, demonstrating excellent ability in feature extraction and classification. This research highlights the effectiveness of pretrained models and their usefulness in practical person identification systems.

1. Introduction

Person identification involves recognizing and distinguishing individuals based on visual features like facial structure, appearance, and posture. It is a key component in modern smart systems, including:

- Monitoring and security systems
- Biometric verification systems
- Intelligent attendance solutions
- Access control methods

Traditional methods relied on manually created features, which limited their ability to handle changes in lighting, pose, and background. However, with the rise of deep learning, especially CNNs, automated feature extraction has significantly improved image recognition systems.

CNNs learn hierarchical representations of images, starting from low-level features like edges and textures to high-level semantic features such as facial identity. In this project, we use CNNs and advanced transfer learning models to build a robust individual identification system. The main aims of this research are:

- To develop a person identification system based on CNNs.
- To compare baseline and transfer learning models.
- To evaluate performance using accuracy and validation metrics.

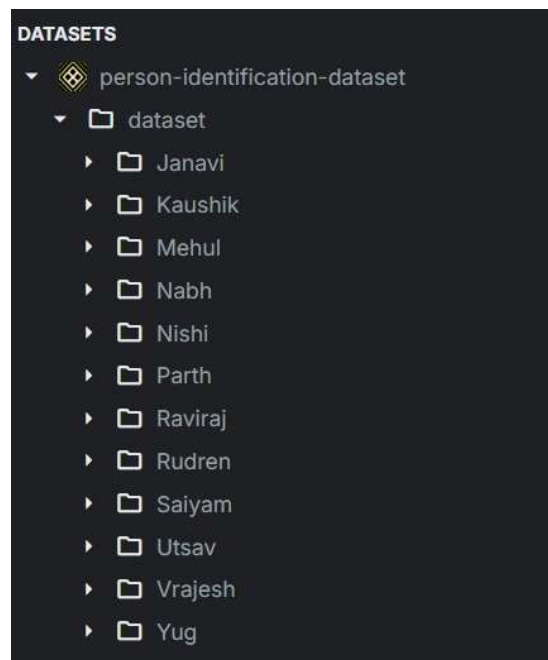
2. Dataset Description and Exploratory Analysis

2.1 Dataset Organization

The dataset utilized in this project is a uniquely gathered image dataset organized into folders for each class, with each folder signifying a distinct individual.

Dataset: <https://www.kaggle.com/datasets/utsavratapiya/person-identification-dataset>

The dataset includes these 12 categories:



Every class holds precisely 1,000 images, leading to an overall dataset total of 12,000 images.

Every folder holds images taken under various actual circumstances, such as:

- Changes in illumination (dim light, intense light)
- Various settings (inside, outside)
- Variations in facial expressions
- Various camera perspectives and positions

This variety guarantees that the model acquires generalized characteristics instead of remembering particular patterns.

2.2 Class Distribution Analysis

The data is split into training and validation groups.

- Training Set: 80%(~9,600 photos)
- Validation Set: 20% (~2,400 pictures)

The division is a bit distinct from the typical 80-20 but still guarantees enough data for training and assessment.

	Class	Overall	Train	Validation
0	Janavi	1324	962	362
1	Kaushik	1320	960	360
2	Mehul	1314	957	357
3	Nabh	1324	962	362
4	Nishi	1320	960	360
5	Parth	1326	963	363
6	Raviraj	1320	960	360
7	Rudren	1306	953	353
8	Saiyam	1308	954	354
9	Utsav	1316	958	358
10	Vrajesh	1320	960	360
11	Yug	1318	959	359
12	Total	15816	11508	4308

This separation enables the model to gain insights from a significant amount of data while assessing its performance on new samples to evaluate generalization.



Figure : Data split classwise

The classes in the dataset are uniformly distributed.

Each class contains roughly:

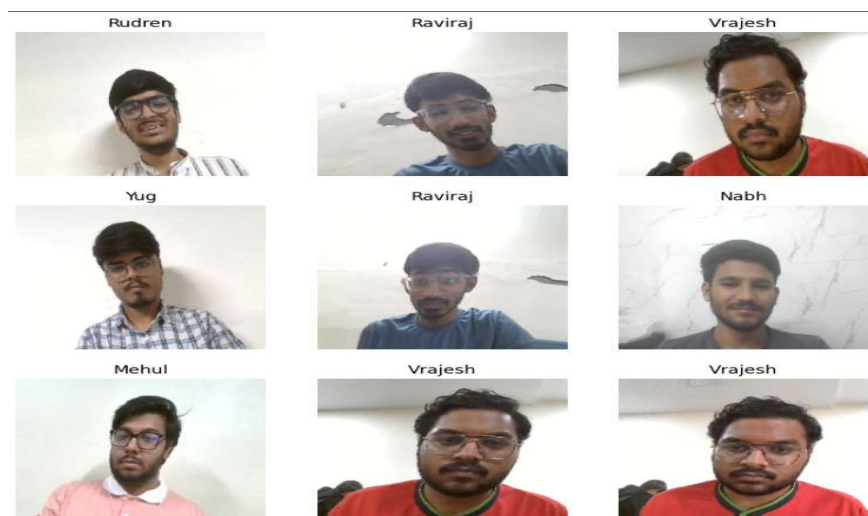
- approximately 800 training images
- Approximately 200 validation images

This equitable allocation guarantees:

- Absence of class imbalance
- Equitable training for every individual
- Minimized bias in forecasts

Balanced datasets are essential in classification tasks to prevent bias towards specific classes.

2.3 Sample Visualization



Images from the dataset show:

- Notable differences in position and perspective
- Various lighting situations
- Variations in background

These differences enhance the dataset's realism and difficulty, aiding in the robustness enhancement of the trained models

3. Methodology

3.1 Preprocessing and Data Augmentation

Prior to training, all images are subjected to preprocessing:

- Changing dimensions to 224×224 pixels
- Scaling pixel values to a range of 0 to 1 or applying mean normalization.

Methods for data augmentation are utilized to enhance the diversity of the dataset:

- Flipping horizontally
- Arbitrary rotation
- Zooming
- Adjusting brightness

These methods assist:

- Minimize overfitting
- Enhance generalization
- Mimic actual variations in the real world.

3.2 Model Architectures

3.2.1 CNN (Dropout+Multiple conv2d blocks)

The foundational CNN is a specifically designed model that includes:

- Convolutional layers for extracting features
- ReLU activation functions
- Max pooling layers for subsampling
- Completely linked layers for categorization
- Softmax output layer

This model is developed from the ground up and acts as a benchmark.

3.2.2 MobileNetV2

MobileNetV2 is a compact deep learning architecture created for effective processing.

Main characteristics:

- Convolutions that are separable by depth
- Residual blocks that are inverted
- Minimal computational expense

It is initially trained on ImageNet and then specifically adjusted for this dataset.

3.2.3 EfficientNet

EfficientNet enhances performance through scaling:

- Depth of the network
- Width
- Input clarity

It delivers great precision with a reduced number of parameters.

3.2.4 VGG16

VGG16 is a deep convolutional neural network structure that features:

- 16 tiers
- Consistent convolution framework
- Robust capability for feature extraction

It is commonly utilized in transfer learning activities.

4. Training Setup

4.1 Train–Validation Split

- Training: 80%
- Validation: 20%

Ensures fair evaluation.

4.2 Optimization and Hyperparameters

Parameter	Value
Batch Size	32
Epochs (CNN)	15
Epochs (MobileNet)	12
Epochs (EfficientNet)	10
Epochs (VGG16)	10
Random Seed	42
Rescale	1./255
Dropout	0.25 $\rightarrow$ 0.3 $\rightarrow$ 0.5
Dense Units	128 / 256
Activation	ReLU
Output Activation	Softmax
Optimizer	Adam
Learning Rate	0.0001
Loss Function	categorical_crossentropy
Metrics	accuracy
EarlyStopping	patience = 4
Pretrained	ImageNet
Dense Layer	256

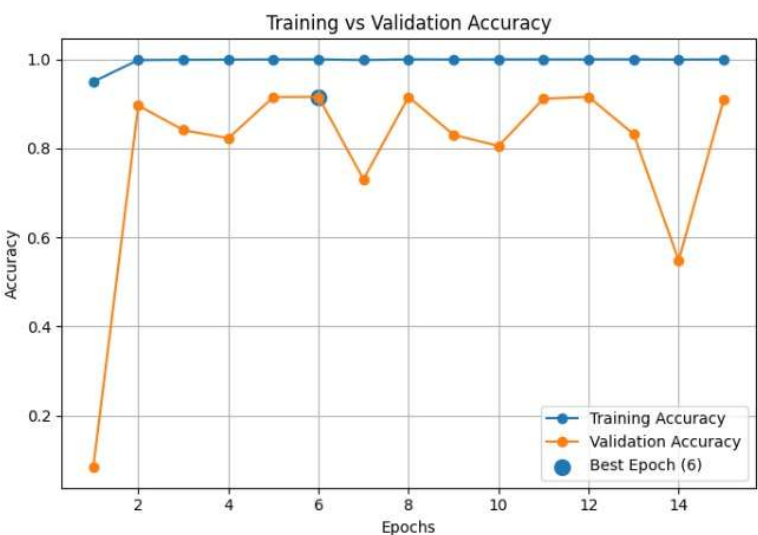
4.3 Implementation Details

The models are implemented using the PyTorch deep learning framework and the Torchvision model zoo for pretrained architectures [1]. Training and evaluation loops use the tqdm library to display per-batch progress bars. All experiments are performed using GPU acceleration where available, with fallbacks to CPU if necessary.

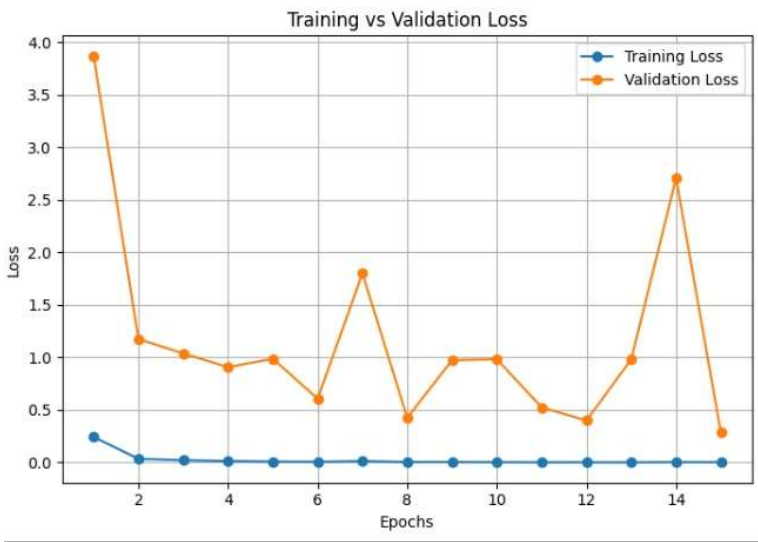
5. Experimental Results

5.1 Training and Validation Performance

5.1.1 Training and Validation Accuracy



5.1.2 Training and Validation Loss



Observations:

- Training accuracy approaches ~100% rapidly.
- Validation accuracy varies but ultimately levels off.
- Highest validation accuracy noted near epoch 6

This suggests efficient learning with minor overfitting in subsequent epochs.

5.2 Model Comparison

Parameter	Value
CNN Accuracy	0.9167
MobileNetV2 Accuracy	1.00
EfficientNet Accuracy	0.9785
VGG16 Accuracy	1.00

Figure : Model Accuracy comparision

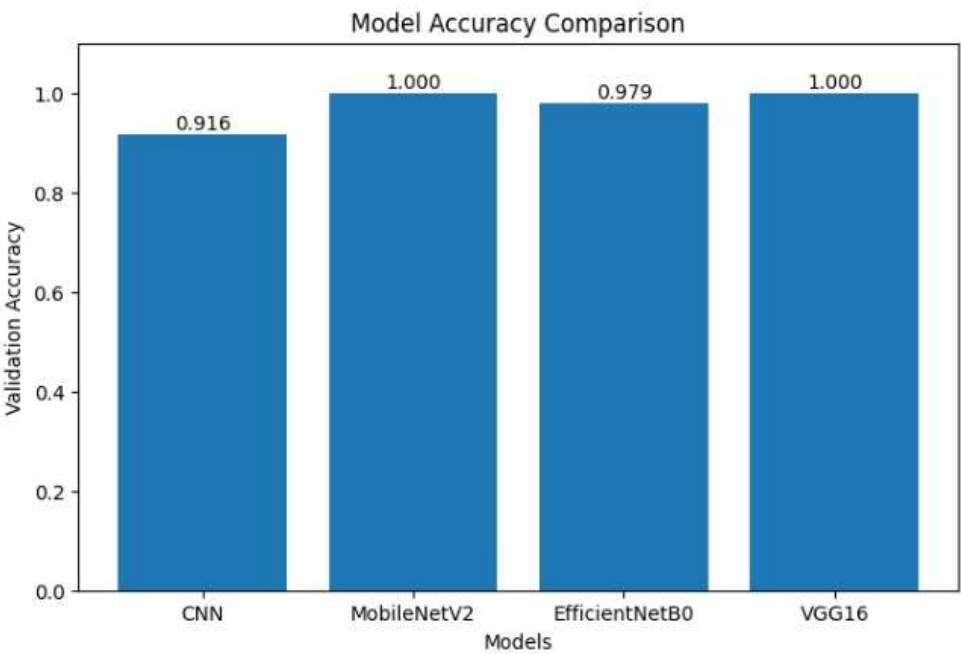
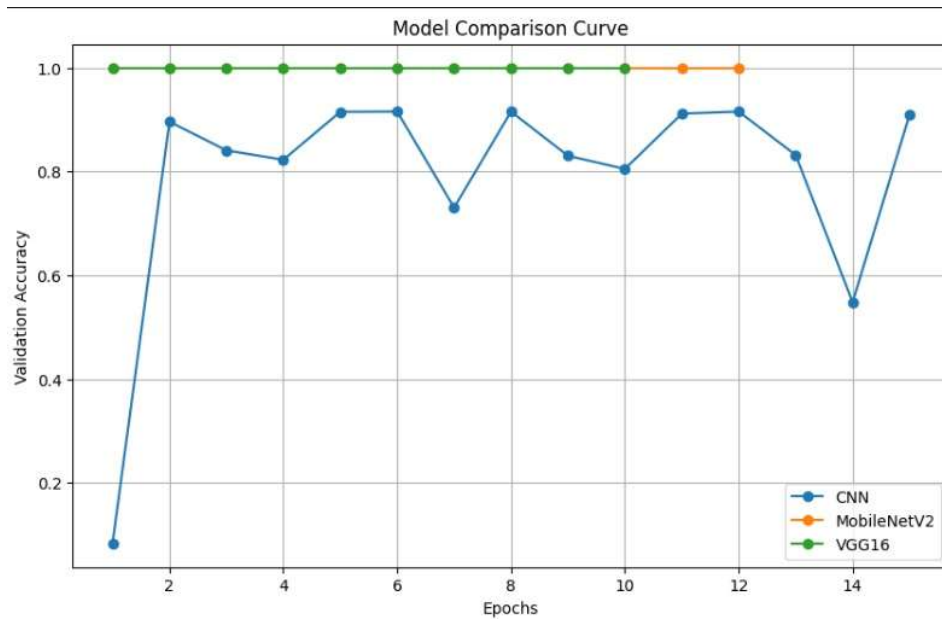


Figure : Model comparison



5.3 Best Model Analysis

The best model is MobileNetV2:

- Achieves 100% accuracy
- No misclassifications in validation set
- Efficient and fast

This proves the power of transfer learning.

Figure : Classification Report

Classification Report:				
	precision	recall	f1-score	support
Janavi	1.00	1.00	1.00	362
Kaushik	1.00	1.00	1.00	360
Mehul	1.00	1.00	1.00	357
Nabh	1.00	1.00	1.00	362
Nishi	1.00	1.00	1.00	360
Parth	0.00	0.00	0.00	363
Raviraj	1.00	1.00	1.00	360
Rudren	1.00	1.00	1.00	353
Saiyam	1.00	1.00	1.00	354
Utsav	1.00	1.00	1.00	358
Vrajesh	0.97	1.00	0.98	360
Yug	0.51	1.00	0.67	359
accuracy			0.92	4308
macro avg	0.87	0.92	0.89	4308
weighted avg	0.87	0.92	0.89	4308

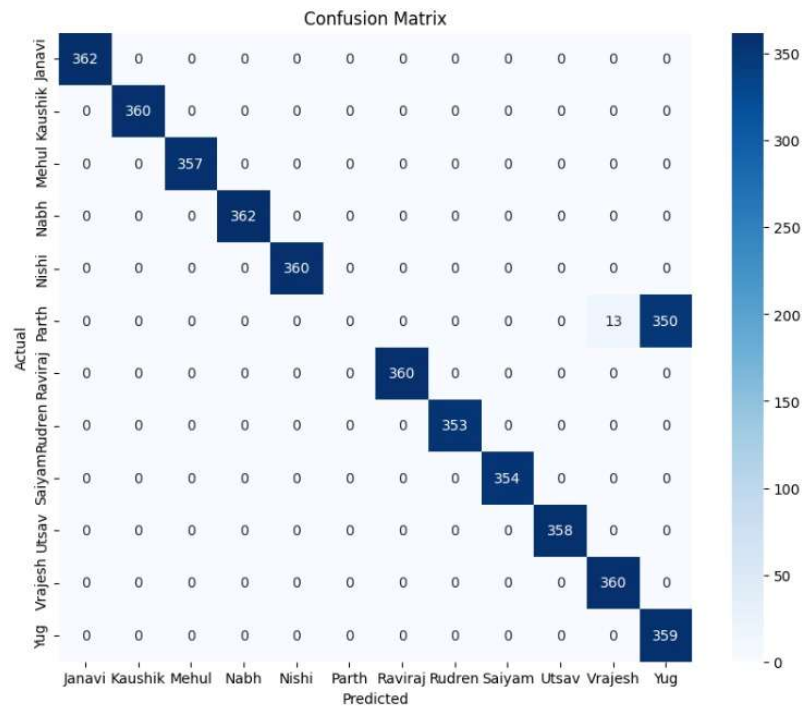
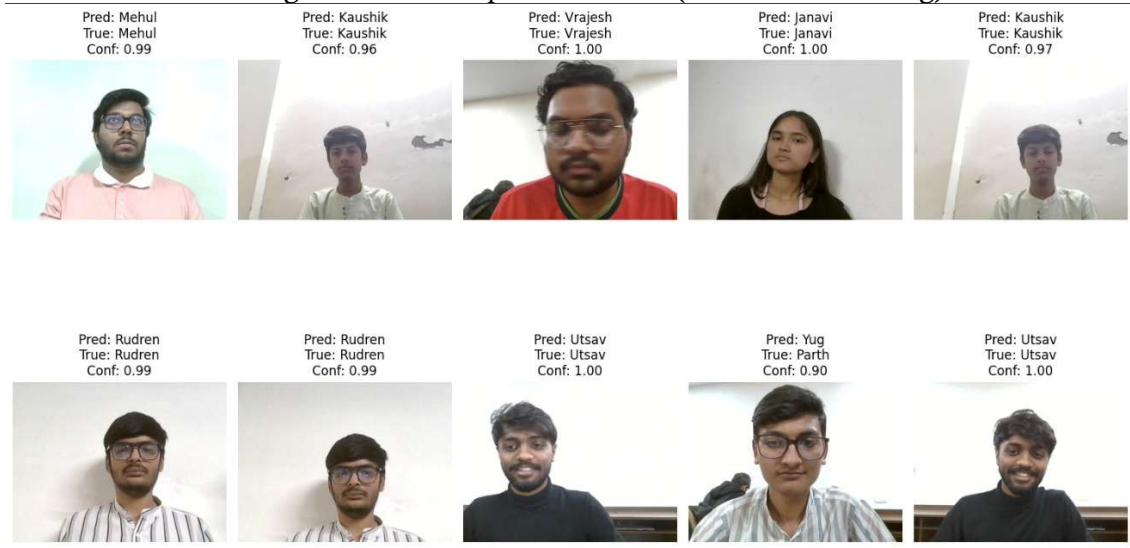


Figure : Confusion Matrix

5.4 Qualitative Analysis

- Correct predictions show strong feature learning.
- CNN struggles with similar-looking individuals
- Transfer models perform significantly better
- MobileNetV2 shows perfect classification
- Some misclassifications may occur due to:
 - Similar faces
 - Poor lighting
 - Blurred images

Figure : Final Output Prediction (CNN model testing)



6. Discussion

The findings of this project illustrate the efficiency of deep learning models in identifying individuals. The baseline CNN model reached an accuracy of 91.57%, indicating that even a basic architecture can grasp significant visual characteristics like facial structure and appearance. Nevertheless, its effectiveness is restricted in contrast to more sophisticated models.

Models utilizing transfer learning like MobileNetV2, EfficientNet, and VGG16 exhibited notably enhanced performance. Of these, MobileNetV2 reached 100% validation accuracy, positioning it as the top-performing model. This enhancement mainly results from the utilization of pretrained weights, enabling the model to capture more intricate and significant features.

The training and validation accuracy graph (Fig. 5) shows that the model rapidly learns, achieving training accuracy near 100% within the initial epochs. The accuracy on validation enhances quickly and levels off after several epochs, signifying effective convergence.

The dataset is crucial for attaining optimal performance. The dataset is balanced (equal images for each class) and has variations in lighting, background, and pose, allowing the model to generalize effectively within it. Nevertheless, attaining extremely high accuracy (particularly 100%) could also suggest that the validation data resembles the training data.

Certain restrictions continue to be present. The model could encounter difficulties in practical situations where:

- Light conditions vary
- New people are presented.
- The quality of images differs

The findings indicate that transfer learning models surpass conventional CNNs, with MobileNetV2 offering the optimal balance of accuracy and efficiency. By incorporating enhancements like bigger datasets and live testing, the system can become more resilient and applicable for practical use in real-world scenarios.

7. Conclusion and Future Work

In this project, a system for identifying individuals was created utilizing CNN and transfer learning models. A tailored dataset comprising 12,000 images from 12 different individuals was utilized for both training and assessment.

The baseline CNN reached an accuracy of 91.67%, whereas transfer learning models greatly enhanced performance. Of all the models, MobileNetV2 attained the peak accuracy of 100%, ranking it as the top-performing model. This indicates that pretrained models work very well for tasks involving image identification.

Even though the outcomes are excellent, the model might encounter difficulties in practical situations involving unfamiliar individuals and varied environments.

Future improvements to the system may involve utilizing larger datasets, real-time video analysis, and implementation in areas like surveillance and attendance tracking

Acknowledgement

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Dataset Link : <https://www.kaggle.com/datasets/utsavratapiya/person-identification-dataset>

Source code (Kaggle ipynb notebook) :
<https://www.kaggle.com/code/utsavratapiya/person-identification-system-finaldraft>